PROBLEM CL13

Centrifugal Pump: Exit Velocity, Axial Force, Torque and Power

Original statement · CL13.pdf

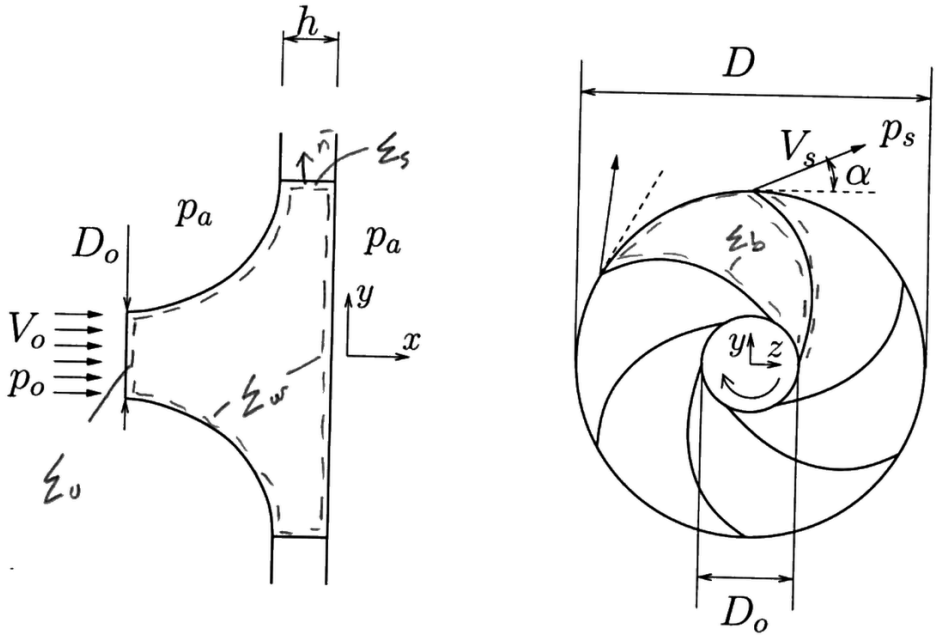
Turbomachine | Angular momentum | Euler torque | Pump power

OFFICIAL PROBLEM STATEMENT

The figure of CL13.pdf shows a sketch of a centrifugal pump. Water enters the pump with uniform axial velocity V_o and pressure p_o through a circular pipe of diameter D_o. The internal geometry of the pump includes rotating blades that redirect the flow, so that the water leaves the pump with zero axial velocity through a lateral exit of width h and diameter D. At the exit section, the pressure is $p_s \neq p_a$ and the velocity vector, of magnitude V_s, forms an angle α with the lateral surface, as indicated in the figure, i.e., the corresponding normal and tangential velocity components are $V_s \sin\alpha$ and $V_s \cos\alpha$, respectively. Assuming that the effect of gravity is negligible and that the water temperature at the exit is equal to that found at the entrance (handwritten addition: and that the pump walls are insulated), determine the values of the:

- Radial velocity component at the exit section $V_s \sin\alpha$.
- Axial force exerted on the pump.
- Torque exerted on the pump.
- Power of the pump, assumed to be ideal (i.e., work per unit time exerted on the water stream by the rotating parts).

Note: besides the water density ρ and the ambient pressure p_a, the following quantities can be assumed to be known in solving the problem: D, D_o, h, α, V_o, p_o, and p_s.



Statement figure (CL13.pdf, p. 1, cropped): inlet D_o (V_o, p_o), lateral exit of width h and diameter D, exit velocity V_s at the angle α with the lateral surface; the surface names are handwritten on the official sheet

`Source: CL13.pdf (page 1, exam of Jan 10, 2012). Notes.pdf, Chapter 3: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.`

Worked solution

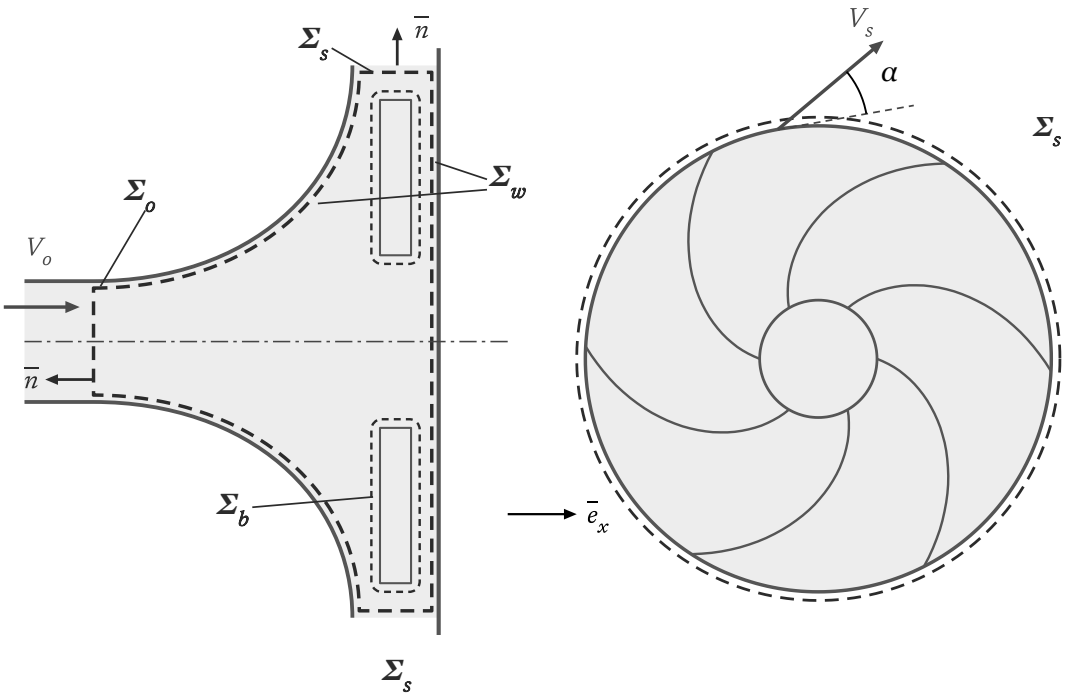
Hypotheses, Data and Unknowns

- Steady (on average), incompressible water; neglect $\bar{g}$; equal temperature at entrance and exit ($e_s = e_o$); insulated walls; uniform properties at both sections.
- The outer casing is surrounded by air at p_a (gauge pressure $p - p_a$ in the force).
- Data: $[\rho,\ p_a,\ D,\ D_o,\ h,\ \alpha,\ V_o,\ p_o,\ p_s]$. Unknowns: $[V_s \sin\alpha,\ F_x,\ \bar{M},\ \dot{W}]$.

Symbol	Meaning	SI unit
D_o, V_o, p_o	inlet diameter, velocity, pressure	m, m/s, Pa
D, h	exit diameter and width	m
V_s, α, p_s	exit velocity, angle, pressure	m/s, rad, Pa
$\dot{W}$	power given to the water by the blades	W

Frames, Control Volume and Surfaces

- Frame fixed to the casing; $\bar{e}_x$ along the axis (inlet flow direction); cylindrical basis $(\bar{e}_r, \bar{e}_\theta)$ around $\bar{e}_x$ with $\bar{e}_r \wedge \bar{e}_\theta = \bar{e}_x$; origin on the axis in the exit plane. Over a full turn the integrals of the unit vectors vanish: with two fixed unit vectors $\bar{e}_y$, $\bar{e}_z$ normal to $\bar{e}_x$ ($\bar{e}_y \wedge \bar{e}_z = \bar{e}_x$), $\bar{e}_r = \cos\theta\,\bar{e}_y + \sin\theta\,\bar{e}_z$ and $\bar{e}_\theta = -\sin\theta\,\bar{e}_y + \cos\theta\,\bar{e}_z$, so $\int_0^{2\pi} \bar{e}_r\,d\theta = [\sin\theta]_0^{2\pi}\,\bar{e}_y + [-\cos\theta]_0^{2\pi}\,\bar{e}_z = \bar{0}$ and $\int_0^{2\pi} \bar{e}_\theta\,d\theta = [\cos\theta]_0^{2\pi}\,\bar{e}_y + [\sin\theta]_0^{2\pi}\,\bar{e}_z = \bar{0}$.
- Surfaces: Σ_o (inlet disc, $\bar{n} = -\bar{e}_x$), Σ_s (lateral exit cylinder, radius $D/2$, width h, $\bar{n} = \bar{e}_r$, $\bar{v} = V_s \sin\alpha\,\bar{e}_r + V_s \cos\alpha\,\bar{e}_\theta$), Σ_w (fixed casing) and Σ_b (rotating blades).



Control volume of the solution: Σ_o (inlet disc, $\bar{n} = -\bar{e}_x$), Σ_s (lateral exit cylinder of radius $D/2$ and width h, $\bar{n} = \bar{e}_r$), the fixed casing Σ_w and the rotating blades Σ_b (schematic); on the right, the exit velocity V_s at the angle α with the lateral surface: normal component $V_s \sin\alpha$, tangential $V_s \cos\alpha$

$$\boxed{\bar{F} = \int_{\Sigma_w, \Sigma_b} \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w, \Sigma_b} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

Balances, Surface by Surface

`Mass (Notes Eq. 3.10)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \bar{v}_c \right) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(\bar{v} - \bar{v}_c \right) \cdot \bar{n} \, d\sigma$
Σ_o	$-\rho V_o \frac{\pi D_o^2}{4}$
Σ_s	$+\rho V_s \sin\alpha \, \pi D h$
$\Sigma_w,\ \Sigma_b$	0

$$\boxed{V_s \sin\alpha = \frac{D_o^2}{4Dh} V_o}, \qquad \dot{m} = \rho V_o \frac{\pi D_o^2}{4} \tag{1}$$

`Linear momentum, axial component (Notes Eq. 3.35)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \bar{v} \, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \bar{v} \left[\left(\bar{v} - \bar{v}_c \right) \cdot \bar{n} \right] d\sigma = - \int_{\Sigma_c} (p - p_a)\, \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma + \underbrace{\cancel{\int_{V_c} \rho \bar{g} \, dV}}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_x \left[(\bar{v}-\bar{v}_c)\cdot\bar{n}\right] d\sigma$	$-\int (p-p_a)\, n_x \, d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x \, d\sigma$
Σ_o	$-\rho V_o^2 \frac{\pi D_o^2}{4}$	$+(p_o-p_a)\frac{\pi D_o^2}{4}$	0
Σ_s	0 ($v_x = 0$)	0 ($n_x = 0$)	0
$\Sigma_w,\ \Sigma_b$	0	$-\int (p-p_a)\, n_x \, d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x \, d\sigma$

Boxed terms: $-F_x$. The radial and azimuthal components vanish because $\int \bar{e}_r \, d\theta = \int \bar{e}_\theta \, d\theta = 0$.

$$\boxed{\bar{F} = \frac{\pi D_o^2}{4}\left[\rho V_o^2 + (p_o - p_a)\right]\bar{e}_x} \tag{2}$$

`Angular momentum about the axis (Notes Eq. 3.38)`

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\, \bar{x}\wedge\bar{v}\, dV}}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\, d\sigma = -\int_{\Sigma_c} \bar{x}\wedge(p-p_a)\bar{n}\, d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})_x \left[(\bar{v}-\bar{v}_c)\cdot\bar{n}\right] d\sigma$	pressure and viscous moments
Σ_o	0 ($\bar{v} \parallel \bar{e}_x$, no swirl)	0
Σ_s	$\rho V_s \sin\alpha\, \pi D h \frac{D}{2} V_s \cos\alpha = \dot{m}\frac{D}{2} V_s \cos\alpha$	0 ($\bar{x}\wedge\bar{e}_r$ has no x component)
$\Sigma_w,\ \Sigma_b$	0	$-\bar{M}\cdot\bar{e}_x$

On Σ_s, $\bar{x} = \frac{D}{2}\bar{e}_r + x\bar{e}_x$, and the x component of $\bar{x}\wedge\bar{v}$ is $\frac{D}{2}V_s\cos\alpha$ (from $\bar{e}_r\wedge\bar{e}_\theta = \bar{e}_x$). The mass flux in front of it is (1).

$$\boxed{\bar{M} = -\rho V_o \frac{\pi D_o^2}{4} V_s \cos\alpha \frac{D}{2}\bar{e}_x} \tag{3}$$

Total energy (Notes Eq. 3.46; $Q_R \equiv Q_c + Q_r$ is the volumetric heat release)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\left(e+\frac{v^2}{2}\right) dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e+\frac{v^2}{2}\right)\left[(\bar{v}-\bar{v}_c)\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\, d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma + \underbrace{\cancel{\int_{V_c} \rho\bar{g}\cdot\bar{v}\, dV}}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\, d\sigma + \underbrace{\cancel{\int_{V_c} Q_R\, dV}}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\, d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	$-\int \bar{q}\cdot\bar{n}\, d\sigma$
Σ_o	$-\dot{m}\left(e_o+\frac{V_o^2}{2}\right)$	$+p_o V_o \frac{\pi D_o^2}{4}$	0	0
Σ_s	$+\dot{m}\left(e_s+\frac{V_s^2}{2}\right)$	$-p_s V_s \sin\alpha\, \pi D h$	0	0
Σ_b	0	$-\int_{\Sigma_b} p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int_{\Sigma_b} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	0
Σ_w	0	0	0	0 (insulated)

Boxed Σ_b terms: the power $\dot{W}$ of the blades. With $e_s = e_o$ and (1):

$$\boxed{\dot{W} = \rho V_o \frac{\pi D_o^2}{4}\left[\left(\frac{p_s}{\rho}+\frac{V_s^2}{2}\right)-\left(\frac{p_o}{\rho}+\frac{V_o^2}{2}\right)\right]} \quad (4)$$

Interpretation, Limits and Dimensional Check

- (3) is the reaction to the angular momentum given to the water: the blades give the water the swirl $V_s\cos\alpha$ at radius $D/2$, and the water acts on the pump with the opposite torque (Euler's turbomachinery relation; the motor must supply it).
- (4): the ideal power raises the total pressure $p+\frac{1}{2}\rho V^2$ of the water flow rate $\dot{m}/\rho$.
- Dimensions (SI): $[\dot{m}\, D\, V_s] = \mathrm{N\cdot m}$; $[\dot{m}\, p/\rho] = \mathrm{W}$; $[D_o^2 V_o/(Dh)] = \mathrm{m/s}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $D = 0.3$ m, $D_o = 0.1$ m, $h = 0.02$ m, $\alpha = 30^\circ$, $V_o = 3$ m/s, $p_o = 1.2\times10^5$ Pa, $p_s = 2.5\times10^5$ Pa, $p_a = 101325$ Pa give $V_s\sin\alpha = 1.25$ m/s ($V_s = 2.5$ m/s), $F_x = 217.36$ N, $|\bar{M}| = 7.652$ N·m, $\dot{W} = 3030.7$ W.
- Agreement with the official solution: CL13.pdf obtains (1) to (4).